ASIM POUDEL

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Portfolio | LinkedIn | GitHub

PROFILE

Final-year Computer Engineering student with a strong foundation in core computer science and hands-on experience building end-to-end systems. Developing machine learning and full-stack applications to solve practical technical problems. Perpetual learner focused on translating requirements into functional, production-ready solutions.

EDUCATION

Kathmandu University **Jan 2023-Jan 2027 (Expected)**
Bachelor of Engineering in Computer Engineering *Dhulikhel, Nepal* | CGPA: **3.87 / 4**

Relevant Coursework: Data Structures & Algorithms, Neural Networks & Deep Learning, Artificial Intelligence, Software Engineering, Database Management, System Analysis & Design, Computer Architecture, Data Mining, Machine Learning, Internet of Things

SKILLS

Languages: Python, TypeScript, JavaScript, HTML/CSS, C++, C, SQL
Frameworks & Libraries: React.js, Next.js, FastAPI, Tailwind CSS, Node.js
AI/ML: LLM APIs, RAG, Vector Embeddings, PyTorch, TensorFlow, scikit-learn, Computer Vision, Sentence-Transformers
Data: NumPy, Pandas, Matplotlib, Feature Engineering
Databases: PostgreSQL (Supabase/pgvector), MongoDB, MySQL, Firebase, SQLite
Tools & DevOps: Git, GitHub, GitHub Actions, Docker, Vercel

PROJECTS

- **Global Commodity Trade Analytics** | *Python, FastAPI, React, Pandas, scikit-learn, statsmodels* *2026*
  - Built a React/FastAPI analytics platform for a 1.2 GB UN Comtrade dataset spanning 1988-2016, supporting country, commodity, trade-flow, regional, trade-balance, growth, and historical trend analysis.
  - Added ARIMA forecasting, Isolation Forest and Z-score anomaly detection, and Random Forest modeling; applied change-point detection, country clustering, regression, and correlation analysis.
  - Accelerated exploration with Parquet caching and pre-aggregated trade cubes; exposed filters, rankings, drivers, forecasts, and CSV export through REST endpoints.
- **Current Price** | *Python, TensorFlow/Keras, CNN-LSTM, Pandas, SMARD* *2025-2026*
  - Developed a CNN-LSTM day-ahead electricity-price forecasting system for the Germany/Luxembourg market using 2022-2024 SMARD data to predict the next 24 hourly prices in EUR/MWh.
  - Engineered a 168-hour lookback pipeline with price/load variables, cyclical time features, and DST-aware timestamps; benchmarked CNN-LSTM and LSTM models against daily and weekly naive baselines using MAE, RMSE, and sMAPE.
  - Saved reusable preprocessing and model/scaler artifacts for production inference and visualization.
- **Conic Section Visualizer** | *Python, JavaScript, Three.js, Vite, Vercel* *2026*
  - Built an interactive 3D visualizer that computes plane/cone intersections and classifies the result as a circle, ellipse, parabola, hyperbola, or degenerate conic.
  - Implemented a Python geometry API and Three.js interface with adjustable cone geometry, conic presets, orbit controls, and real-time rendering.
  - Added regression tests for cone dimensions, clipping, classification boundaries, and invalid inputs; deployed the Vite frontend and Python API on Vercel.
- **Skill Forge** | *React, FastAPI, PyTorch, scikit-learn, SQLAlchemy* *2026*
  - Built a gamified adaptive learning platform with ML-powered learning-style personalization (decision tree + PyTorch MLP) driving quiz difficulty and skill recommendations.

RESEARCH

- **Rx Pilot** | *Multi-Model OCR Pipeline, Custom Deep-Learning Model* *In Progress*
    - Developing an OCR pipeline for handwritten prescription digitization by benchmarking state-of-the-art models against a custom-trained architecture.
    - Building a reliable ground-truth dataset through data collection, manual annotation, and quality-controlled labeling.

LEADERSHIP / EXTRACURRICULAR

- **Logistics Co-Lead,** Hackathon KU IT Meet (2025)
- **Event Manager,** IT MEET Gaming Competition (2024)
- **Logistics Member,** Hackathon event KU IT Meet (2024)
- **Volunteer,** IT Meet (2023)